SUCC - Sequence and UMI Clustering-based Correction

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Motivation

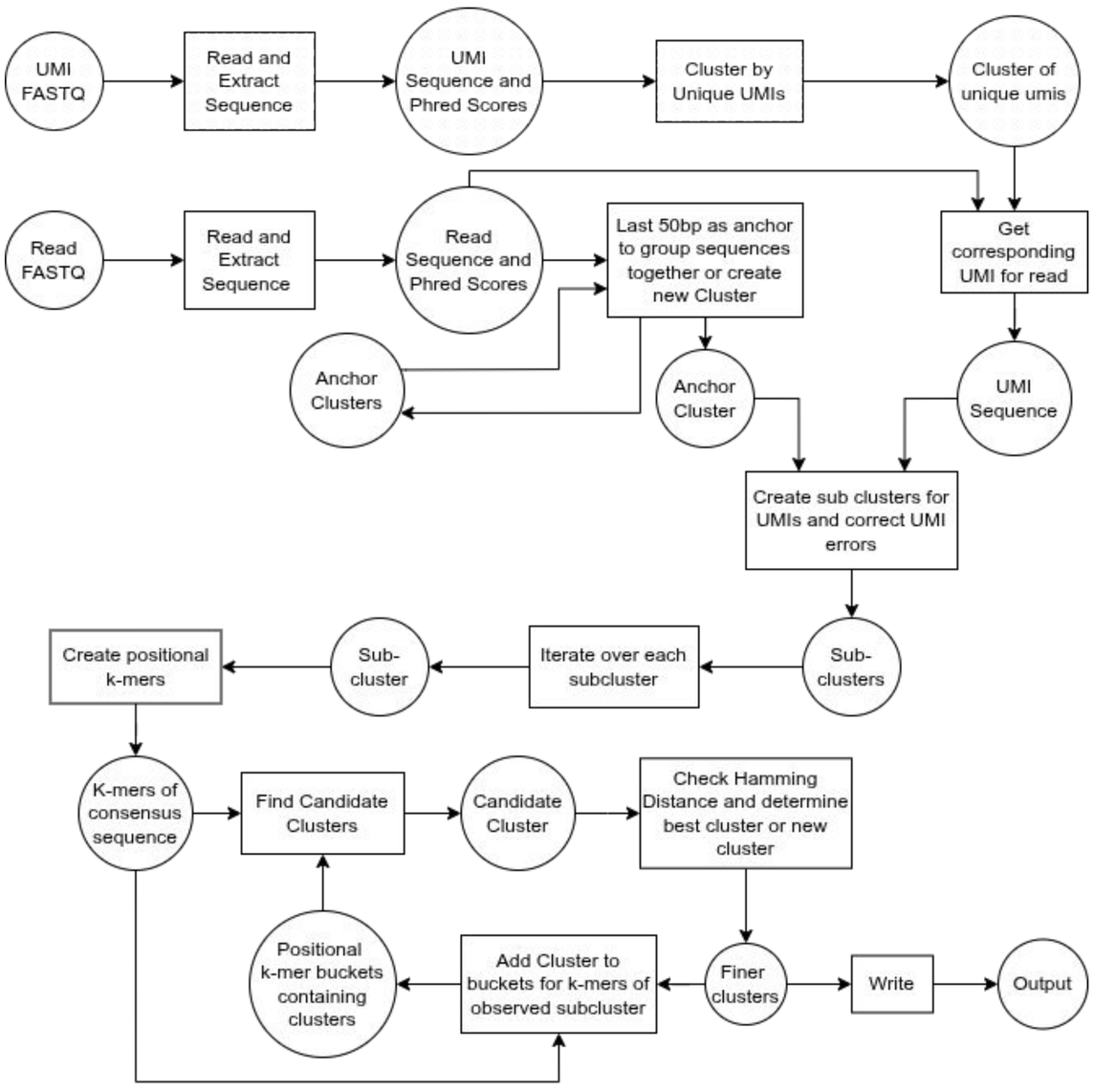
- PCR amplification in RNA-seq causes dis-proportionate duplication of fragments
- Leads to false counts of gene expression
- Unnecessary workload for the mapper

Approach

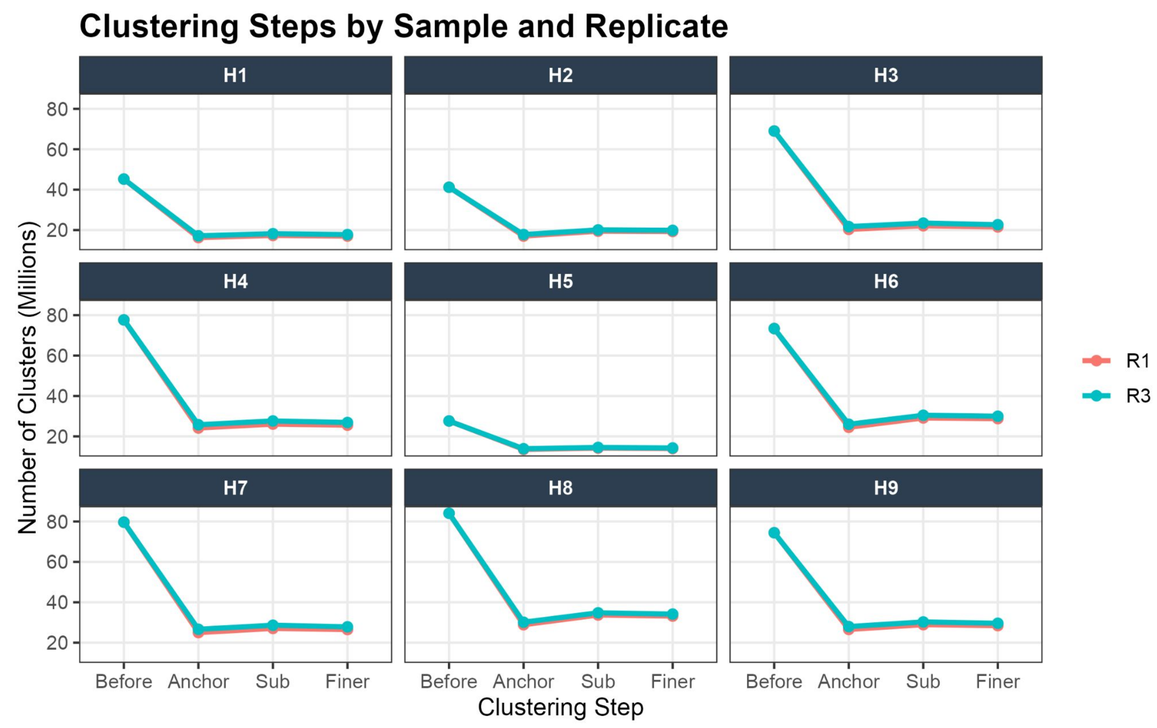
- Unique molecular identifiers are used to label original molecules prior to amplification

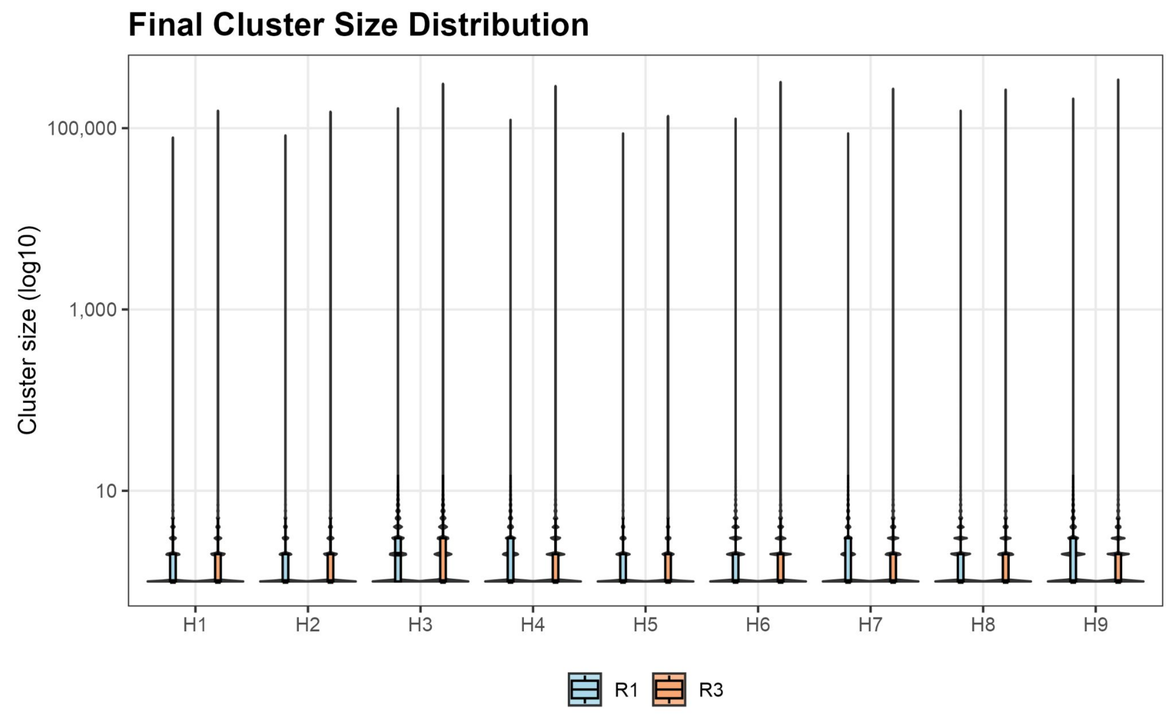
➔ Cluster UMIs and reads together to deduplicate and correct reads and UMIs

Workflow



Results





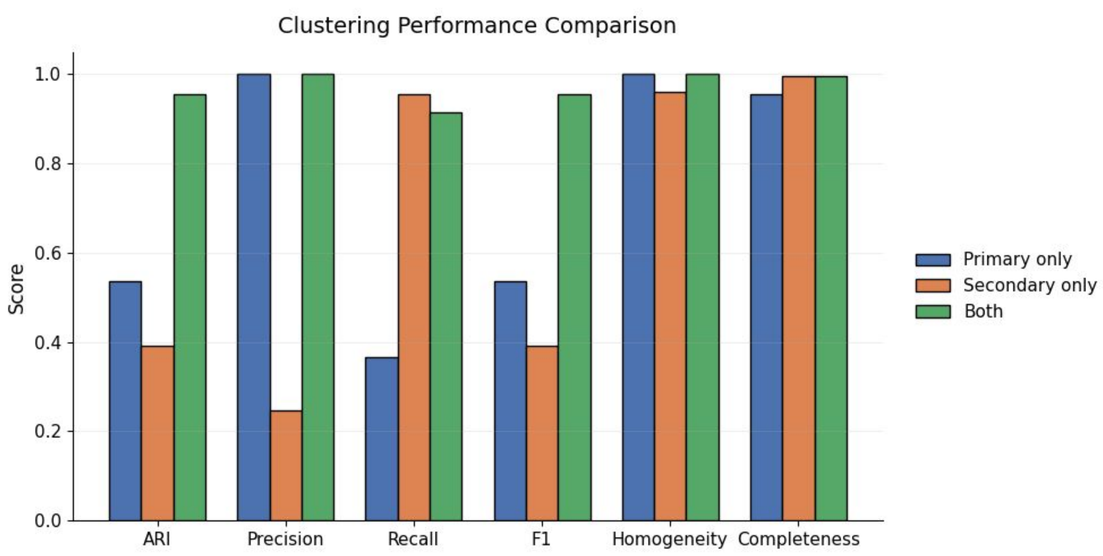
➔ Less than half the amount of reads across all samples of Sus Scrofa RNA-seq datasets.

➔ Most clusters consist of singletons, with a small number of clusters containing a large number of reads

➔ Differing number of clusters between forward and reverse reads

Simulation

- The algorithm was evaluated using a simulated RNA-seq dataset
- The input transcript counts were derived from the Sus Scrofa expression data
- Result were evaluated in 3 modes:
  - Primary only (Anchor based)
  - Secondary only (Finer clustering)
  - Both



- The combination of both clustering methods yield the best clustering performance

Conclusion

- Real data insights:
  - PCR amplification biases and “jackpot” instances impact clustering
  - Forward and reverse read yield different number of clusters

➔ Optimal parameters are dataset-dependent

➔ Careful tuning is required for real word applications